

# Understanding people. Building for real life.



## XINYU WENG

Researcher × AI Product Manager

### Zhengzhou University B.S. in Packaging Engineering

- Overall Rank: Top 5%
- Awarded in 10+ National competitions
- 4-years University Merit Scholarship
- Founded a photography studio, making profit of 10K+ RMB per month.

### South China University of Technology M.S. in Industrial Design Engineering

- National Scholarship (Top 1% for academic excellence)
- Practice-Oriented Graduate Student Honor (1%)
- Published 2 papers in top SSCI journals (Q1 & Q3)
- Master's thesis received Double-A (Top 5%)

### BYD Auto Industry Co., Ltd. Senior Product Manager

- Recognized as High-Potential Talent
- Independently took charge of the charging app, boosting its usage rate from 13% to 30%, and UX scores from 5/10 to 8/10
- Conduct on-site research in Manchester for localized design

### Wangsu Technology Co., Ltd. Senior Product Manager

- Led the 0-to-1 plan and design of SalesAI (AI Assistant app)
- Constructed domain-specific agent workflows, and optimized LLM intent recognition strategy

Affective Computing  
Immersive HCI (AR)  
Digital Health  
Quantitative Experimental Design

C-End Product Thinking  
Global Vision  
User Growth

B-End Product Thinking  
AI Product Thinking  
Agent Implementation

Sep. 2016 - Jun. 2020

Design Awakening

Sep. 2020 - Jun. 2023

Research

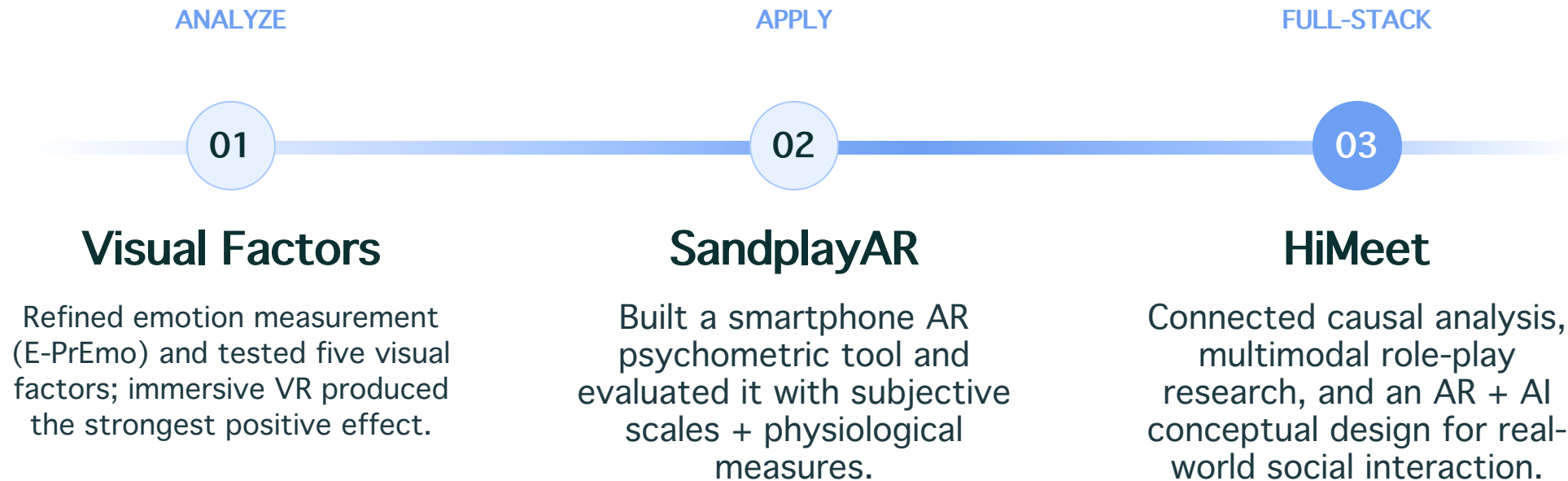
July 2023 - July 2025

Industry Practice

July 2025 - Present

Ph.D

**One consistent thread: multimodal measure human states, then translate evidence into interaction.**



**Measurement innovation → multimodal validation → evidence-led design**

# Visual factors shape distinct positive emotions—not just generic valence.

Research on the Influence of Visual Factors on Emotion Regulation Interaction

## PROBLEM & GOAL

Digital products are accessible emotion-regulation tools, yet there was no clear consensus on how specific visual elements influence complex emotions. The study tested five factors and sought design-ready guidance.

## METHOD

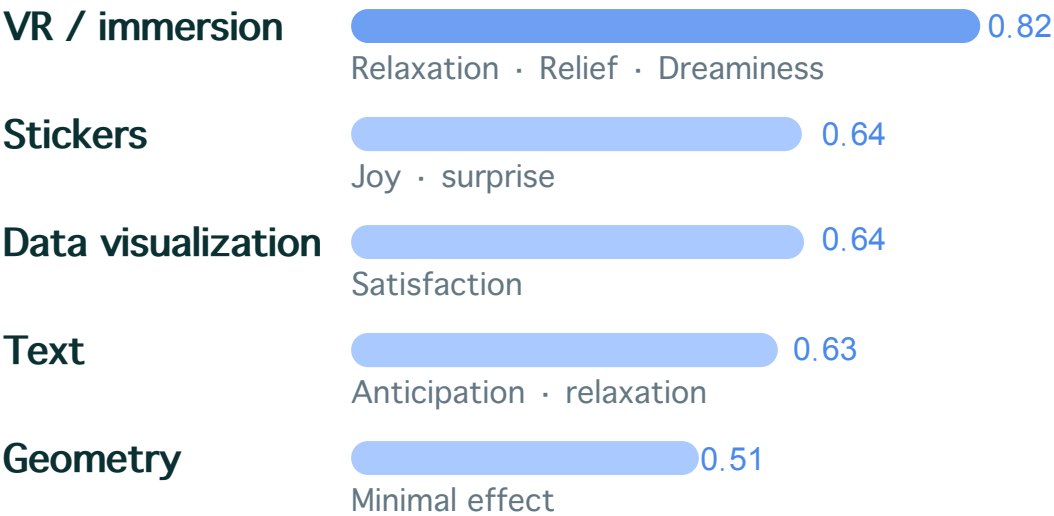
[More detail ->](#)

- E-PrEmo: 10 context-specific emotions selected from a 60-emotion pool;
- Quantification: rank-weighted score  $10(n - m + 1)/n$ ;
- Experiment: within-participant(10 Asian young adults), randomized five-factor exposure, ranked the E-PrEmo cards.

## MAIN RESULT

- All five factors increased positive emotion.
- VR > Stickers >Data viz >Text >Geometry

## OBSERVED EFFECT RANKING



## NOW I WOULD IMPROVE

- More emotion stimulation
- Larger and culturally diverse sample
- Stronger Control Conditions

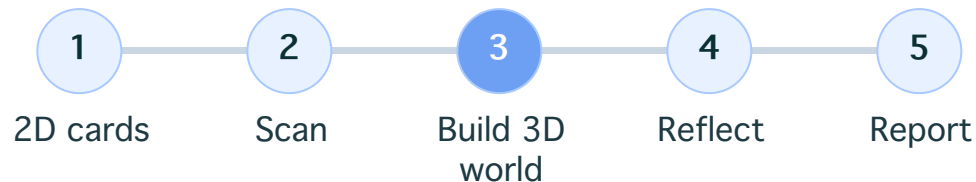
# SandplayAR moved a material-intensive therapeutic practice onto a smartphone.

Evaluation of psychometric game for people with generalized anxiety disorder

## PROBLEM & GOAL

Traditional sandplay depends on physical materials and repeated in-person sessions. Cost, logistics, and stigma limit access. **Goal:** design a portable AR tool and evaluate its experience and calming effects for anxious young people.

## SYSTEM FLOW



## EVALUATION

N = 20 (age 15–30, GAD score >5), AR vs. Traditional sandplay  
SIHEQ + SUS for testing experience, Skin Conductance + Systolic Blood Pressure for testing changes of tension or anxiety.

## MAIN RESULTS

$p < .05$

### Analysis of questionnaire data:

Better unconscious emergence and conscious activation than traditional sandplay.

$p = .023$

### Physiological data analysis:

Skin conductance decreased only in the AR group; SBP decreased in both groups.

71/100

### Usability data analysis:

The system has met the basic usability requirements

## NOW I WOULD IMPROVE

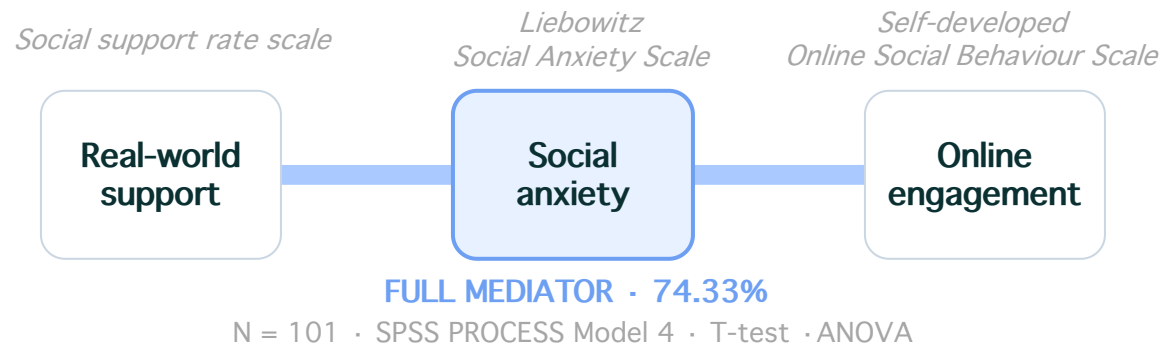
- larger sample;
- optimize system design, not just a rough demo;
- Use multimodal sensing to provide visual biofeedback in system.

# A full-stack programme: explain the phenomenon, locate its triggers, then design support.

## 01 DISCOVERY · CAUSAL MODEL

### Question

- Why do heavy online socialisers tend to have smaller offline social circles and show more social-anxiety-related behaviours?
- Does social anxiety explain the link between weaker real-world support and greater online social engagement?



### Key Findings

- Weaker real-world social support was associated with higher social anxiety and greater dependence on online socialization.
- Young adults aged 18–25 were especially vulnerable

**Design implication:** digital support should reconnect people with real-world relationships.

## 02 PERSONAL SAMPLING · ROLE-PLAY test

**Method:** Self-Report + physiological measures+ video observation

- Scripted social scene + LSAS grouping
- **pre-test:** Self-awareness Evaluation Scale; **post-test:** Subjective Units of Distress Scale; Bodily symptoms Scale
- GSR/heart rate + video-coded behavior, aligned by event and time.

**38.0 vs 8.8**

**Exposure scenarios is effective:** socially anxious participants showed significantly higher subjective distress,  $p < .001$

**$Y=2.037*x$**

Social anxiety(Y) **explained(45.1%)** by negative self-cognition(x), especially in negative body-image perceptions, inferiority etc.

### Invention strategies

**Cognitive correction · Skill training · Attention modification · Exposure enhancement · Positive evaluation**

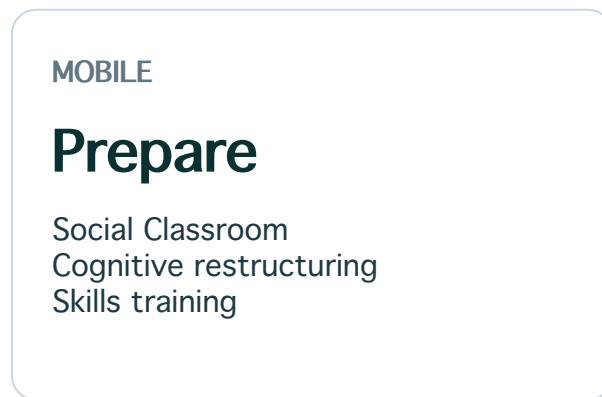
Nevertheless, individuals with varying anxiety levels differ in effect magnitudes. **Interventions should therefore follow general principles alongside personalised tailoring.**

# The solution supports social interaction before, during, and after the real-world encounter.

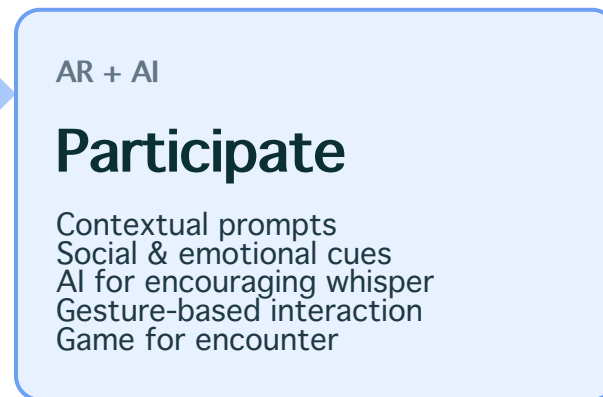
## Core functions

Helping users **relationship initiation**(mitigating start-up barriers); **relationship development**(playful interactions); and **relationship maintenance**( nurture social ties).

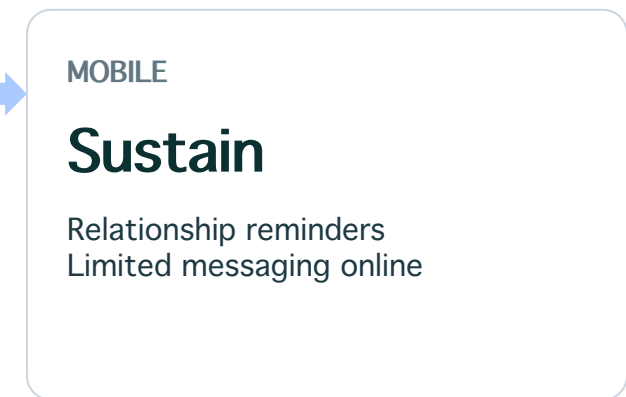
### BEFORE



### DURING



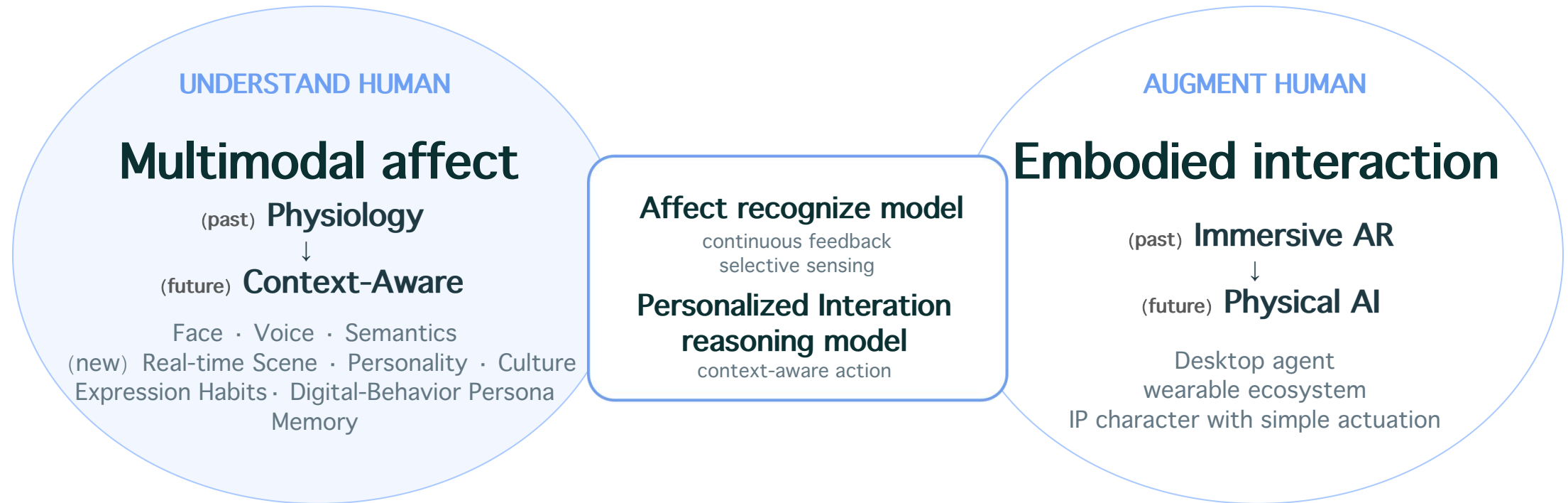
### AFTER



### NOW I WOULD IMPROVE

Develop real and usable programs rather than conceptual solutions; Integrate more AI capabilities, providing personalized suggestions rather than just general strategies via personal data

**My next steps continue to link understanding human affect with real-world assistance.**



The carrier does not have to be humanoid; the research focus is accurate interpretation and appropriate action.

# Affective agents that understand nuance, adapt over time, and act responsibly in real life.

## RESEARCH QUESTIONS

1

### Represent

- Which multimodal features predict individual affect
- Can the model recognize distinct context-dependent meaning, contradiction, and masking?

2

### Adapt

- How should modalities be dynamically weighted and corrected through feedback
- How should predictions guide personalized interaction strategies?

3

### Embody design and responsibly

- How device system combinations across home, work, and mobility for seamless companion purpose;
- How should autonomy, oversight, privacy, and dependence be balanced?

## METHODS

1

### Collect & label

Emotion-induction tasks; diverse culture/personality; multiple sensing +self-report + observer labels.

2

### Model & compare

Mixed-effects screening, adaptive test(dynamic fusion/gating); Mapping recognition+persona with actions through Gamification tool

3

### Validate in life

Participatory design; Cross-device prototype; Long-term user experience evaluation; longitudinal field study; continuous ethics and privacy review.

**Goal: not more sensing—but better interpretation, selective sensing, and more appropriate action.**